B.Tech (5th Semester) Computer Engg, Honors (2025)
DATA ANALYTICS SESSIONAL – 2 (CSH – 511)

Time : 1 Hour **Max. Marks : 15 (4+3+3+5)**

Q.1. Why does partition based algorithm fails to find clusters of arbitrary shapes whereas DBSCAN and OPTICS easily find them?

Q.2. Given the following data points. Picking initial **medoids** as P1, P3 AND P5 show the final clusters formed.

P1 = (2, 6, 4), P2 = (3, 4, 7), P3 = (8, 1, 2), P4 = (7, 2, 3), P5 = (6, 5, 8), P6 = (4, 7, 6)

Q.3. Use single link agglomerative clustering to group the data described by the following distance matrix. Show the dendrograms.
Can we get an idea on the best number of clusters from a dendogram?

	A	B	C	D
A	0	1	4	5
B		0	2	6
C			0	3
D				0

Q.4. How can use DBSCAN algorithm to detect outliers?

Consider the following data points (185, 72), (170, 56), (168, 60), (179, 68), (182, 72), (188, 77), (180, 71), (180,70), (183, 84), (180, 88), (180, 67), (177,76). Assume (185, 72) and (170, 56) as two centroids. Using K-means algorithm show the clusters formed after execution of **2 rounds** of execution. Use Manhattan distance measure.